Python Lab

Instructions:
Attempt all the questions
Use comments and validations wherever it is required.
Use proper name for variables or functions.

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1. Student Grade Analyzer

Create a function calculate_grade(marks) that:

- Takes marks (0–100)
- Uses if-elif-else to assign grades:
  - 90+ → A
  - 75–89 → B
  - 50–74 → C
  - <50 → Fail
- Use a loop to process marks of 5 students

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2. Number Classification System

Write a program that:

- Takes a list of numbers
- Uses a function to check:
  - Positive / Negative / Zero
  - Even / Odd (nested if)
- Use a for loop to process all numbers

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3. Login System (3 Attempts)

Create a login function:

- Username = "admin", Password = "1234"
- Allow max 3 attempts using a while loop
- Use if-else for validation
- Print:
  - "Login Successful"
  - "Account Locked" after 3 failures

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4. Prime Number Checker

Write a function is_prime(n):

- Use for loop to check divisibility
- Return True/False
- Use if-else to display result

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5. Multiplication Table Generator

Create a function:

- Takes a number from user
- Uses for loop to print table (1–10)
- Use if to restrict negative input

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6. Salary Bonus Calculator

Write a function:

- Input: salary and years of experience
- Logic:
  - 10 years → 20% bonus
  - 5–10 years → 10%
  - <5 → 5%
- Use if-elif-else
- Loop for multiple employees

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7. ATM Withdrawal System

Create a function:

- Initial balance = 10000
- Ask withdrawal amount
- Conditions:
  - Insufficient balance
  - Invalid amount (<0)
  - Successful withdrawal
- Use while loop until user exits

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8. Palindrome Checker

Write a function:

- Check if a number/string is palindrome
- Use loop (while or for)
- Use if-else to print result

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9. Pattern Printing with Conditions

Write a program:

- Print triangle pattern using for loop
- If row number is even → print *
- If odd → print #

Example:

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#
* *
# # #
* * * *
```

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10. Menu Driven Calculator

Create a function-based calculator:

- Menu:
  1. Add
  2. Subtract
  3. Multiply
  4. Divide
  5. Exit
- Use while loop for repetition
- Use if-elif-else for operation selection

- Handle divide-by-zero using condition